# Associate Professor Morten Thaysen Andersen, PhD

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***Research interests and expertise areas***

Glycoproteome research, Multi-omics, Glycobiology, Glycoprotein Structure/Function/Biosynthesis, Analytical chemistry, Glycoimmunology, Host/pathogen interactions, Mass spectrometry.

***Research programme***

Andersen's research team (currently 15 members), explores how complex carbohydrates (glycans) in the human glycoproteome affect the function of key proteins in the innate immune system including in inflammation (sepsis), host-pathogen interactions and cancer. His team develops and applies cutting-edge mass spectrometry-based glycomics and glycoproteomics technologies for the large-scale molecular mapping of the heterogenous human glycoproteome and draws on methods in protein and carbohydrate chemistry and molecular immunology to explore structure-function relationships of glycoproteins in immune processes relevant to human health and disease.

***Bio-sketch and overview of contributions to field***

Associate Professor Morten Andersen has 15 years of post-PhD experience. He obtained a PhD degree in Protein Chemistry in 2009 from Univ Southern Denmark (Advisor: Prof Peter Hojrup) and thereafter relocated to Distinguished Prof Nicki Packer's laboratories in Australia to complete two postdoctoral fellowships competitively awarded by the Danish Research Agency and the Australian Research Council (2010-14). Enabled by a competitive Cancer Institute NSW fellowship (2014-17), he then established the Analytical Glycoimmunology lab at Macquarie Univ (MQ). He was awarded an ARC Future Fellowship in 2022 and is CI on the ARC ITTC FAAB on advanced bioactives (2022-25). Recently, Andersen was recruited to the Institute for Glyco-core Research (iGCORE) at Nagoya Univ, Japan as Visiting Professor (20% load, 2022-2027) to set up a clinical glycoproteomics lab as part of a large nationally funded glycoscience program (Human Glycoproteome Atlas, 2023-2033, ~$120M).

A/Prof Morten Andersen has secured as CI more than $12M in career funding since 2014. He is increasingly working with key industry partners including Merck Millipore, Protein Metrics and AusHealth. He has mentored 13 postdocs; 7 postdocs attracted their own competitive fellowships with his support (e.g. Marie Curie European Union, DFG and Cancer Institute NSW). Notably, he has trained 14 HDR students (7 PhDs, 7 MRes students) as a principal supervisor.
Morten is highly research active as documented by his 101 peer-reviewed papers published in trusted and regarded journals including *Nat Methods*, *PNAS*, *Cell Reports*, *MCP*, *Anal Chem*, *JBC* and *Glycobiology* (H-index 44, 5,200+ citations, Google Scholar). He has received 20 awards for scientific achievements including the prestigious Career Development Award 2023 and Ken Mitchelhill Young Investigator Award, 2013 (both from APS) and the Early Career Researcher of the Year – 2013 and FSE Excellence in Research Leadership in 2021 (both at MQ), see below for details.

Morten has contributed to the inter/national community as Director of the Human Glycoproteomics Initiative (HGI) under HUPO since 2020, through conference organisation via the Australian Proteomics Society (2016-21, e.g. organisation of the World Congress HUPO 2019, Adelaide, SA) and peer-review activities for 55+ quality journals (~30 manuscripts/year) and competitive grants for 9 funding bodies including ARC and NHMRC. Morten is the Australian representative for the International Glycoconjugate Organization (a key global community in the scientific field) and he has also shown leadership by being the lead co-founder of the Australian Glycoscience Society (AGS also known as Glyco@Oz) in 2022 with ~130 members. Morten is the inaugural president of the AGS (2022-2025).

***Education and research visits***

- B.Sc. Biochemistry and Molecular Biology, Univ Southern Denmark, 2003
- M.Sc. Biochemistry, Univ Southern Denmark, 2005
- Ph.D. Glycoprotein biology, Univ Southern Denmark, 2009
- Visiting Student, Harvard Medical School (Urology/Proteomics), MA, 2008
- Visiting Research Fellow, Harvard Medical School (Neuroscience/Glycobiology), MA, 2016
- Visiting Researcher, Universität für Bodenkultur Wien (BOKU), Vienna, Australia, 2024 (Invited)

***Employment***

- Danish Research Council-funded Postdoctoral Fellow, Macquarie University (MQ), 2010-12
- ARC Super-Science Research Fellow, Macquarie University, 2012-14
- Cancer Institute NSW-funded Early Career Fellow, Macquarie University, 2014-17
- Career interruption for parental leave (six months@1.0 FTE), 2017-18
- Senior Research Fellow, Macquarie University, 2017-22
- ARC Future Fellow and Associate Professor (tenured, continuing position), 2022-25 (80% load)
- Visiting Professor, iGCORE, Nagoya University, Japan, 2022-27 (20% load)

***Scientific achievements***

*i. Journal editor roles*

- *Glycoconjugate Journal* (Springer, associate editor since 2017)
- *Molecular Omics* (RSC, guest co-editor, special issue, ~20 papers, 2020-21)
- *Essentials of Glycobiology* (4th Ed, CSHL Press, Junior Editor, Chapter 51, 2022)

*ii. Conference contributions* (‡invited/$Keynote or Plenary talk) (selected from ~50 international talks)

- ‡$Thaysen-Andersen. 2023. Society for Glycobiology, Hawaii, US.
- ‡$Thaysen-Andersen. 2023. Glyco26, Taipei, Taiwan.
- ‡$Thaysen-Andersen. 2023. 22nd World Congress HUPO, Busan, South Korea.

*iii. Conference organisation* (selected from ~10 international meetings)

- Conference organiser. 2022. 4th Australasian Glycoscience Symposium, Gold Coast, QLD (140 delegates)
- Symposium chair. 2019. 2nd Australasian Glycoscience Symposium, SA (130 delegates)
- Organising committee (APS-led). 2019. 18th World Congress HUPO, Adelaide, SA (1,000 delegates)
- Symposium chair. 2018. 1st Australasian Glycoscience Symposium, Lorne, VIC (100 delegates)
- Organising committee (APS-led). 2017-21, 22nd-26th Proteomics Symposium, Lorne, VIC (300 delegates)

*iv. Awards and prizes* (Selected from ~20 career and travel awards/prizes)

- Australia Proteomics Society inaugural Career Development Award ($3,000), APS, 2023
- Highly commended, Jim Piper Excellence in Research Leadership (MQ, 2023)
- Winner, Excellent Research Leadership – Science and Engineering (MQ, 2021)
- Mid-career Australia Proteomics Society travel award, kHUPO 2016, Daejeon, Korea, 2016
- Ken Mitchelhill Young Investigator Award ($2,000) (Australasian Proteomics Society, 2013)
- Early Career Researcher of the Year – Science and Engineering ($2,000) (MQ, 2013)

*v. Student training and postdoc mentoring* (only principal supervision listed)

- Trained 7 PhD and 7 Masters of Research students (all on-time completion) since 2015
- Mentored 13 postdocs and ECR fellows since 2014
- Current team size: 15 members (4 postdocs, 7 PhDs, 2 Master students, two volunteers)

*vi. Boards, committees and other service to field (local, national and international)*

- Executive board member, and President, Australian Glycoscience Society (AGS), since 2022
- Executive board member, Australian Proteomics Society (APS), 2016-21
- Australian representative, International Glycoconjugate Organization (IGO) since 2018
- Director and co-founder, Human Glycoproteomics Initiative (HGI), HUPO since 2016
- Co-founder, Glyco@Oz, Australian Glycoscience Network (www.OzGlyco.org), 2020-22
- Expert Witness for global pharmaceutical company since 2023

*vii. Grants (~$12M total career funding as CI since 2014)*

- Three fellowships (~$2M): FNU 2010 ($230k) 2010, CINSW ECF ($700K) 2014, ARC Future Fellowship 2021 ($1M).
- Supported fellowships as mentor (~$2M) for 6 postdocs e.g. Cancer Institute NSW ECF, Marie Curie, DFG, JSPS
- Project/centre funding (~$6M): Awarded project grant funding e.g. ARC ITTC 2021 ($5M)
- Infrastructure (~$2M): Cancer Inst. NSW Instrument ($650k) (CIE) 2015, ($820k) (CIH) 2017.
- Contributed as a Visiting Professor (iGCORE) to securing national Japanese program grant ($120M, 2023-33)
- Recent industry funding: Merck-Millipore ($150K), AusHealth ($180K), Protein Metrics ($80K), Thermo ($100K)

*viii. Reviewer roles*

a) Grant reviewer for ~10 research agencies e.g. NWO, ANR, SNSF, Cancer Research UK, FWF, NHMRC, ARC.
b) Reviewer for 50+ journals: e.g *Chem Rev*, *PLOS Biology*, *Nat Commun*, *Nat Methods*, *Commun Biol*, *E-life,*
c) Examiner of 9 national PhD theses, 2017-23.

*ix. Select papers* (recent key papers)


- Kawahara, Ugonotti, …., **Thaysen-Andersen**. *Proc. Natl. Acad. Sci. U.S.A.,* 120(36)e2303867120, **2023**
- Bagdonaite, .., **Thaysen-Andersen**, and Scott. *Nature Reviews Methods Primers*, 2(48), **2022**.
- Kawahara, Chernykh,…..and **Thaysen-Andersen**. 18(11):1304-1316. *Nat Methods*, **2021**.
- Parker…., **Thaysen-Andersen**, Borrow, Ternette. *Cell Rep*. 35(8):109179, **2021**